SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: ITA 06 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO4: K- Nearest Neighbour Learning – Locally weighted Regression – Radial Bases Functions – Case Based Learning ***(BL 5)***

Assessment Tool Weightage: 20%

QUESTIONS: 03 **Total Marks:25**

Annexure A

Industry Problem Solving

Code of Conduct:

I ***__Bandaru Jayasri-192424301__*** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Criteria	*Conceptual Understanding*	*Accuracy of Answers*	*Application of Knowledge*	*Completeness of Response*	*Clarity & Presentation*	*Total*
Weightage	*25%*	*25%*	*20%*	*15%*	*15%*	
Q1						
Q2						
Q3						

Signature of the student: ***Total Marks :***
B.Jayasri

Annexure A

Short Answer Test

1. A banking system wants to detect potential loan defaulters based on customer financial history and transaction patterns. Design a solution using **K-Nearest Neighbour or Logistic Regression**. Explain how similarity or decision boundaries are used, how features are selected, and how the system adapts to new customer data over time.

2. A cybersecurity firm wants to identify abnormal network behavior to detect intrusions. Design a system using **KNN or Case-Based Learning**. Explain how similarity between network patterns is measured, how anomalies are detected, and how the model updates with new attack patterns.

3. An energy management company aims to predict electricity consumption based on factors such as time, weather, and appliance usage. Propose a solution using **Locally Weighted Regression or RBF networks**. Explain how local learning captures variations in consumption patterns and how the model handles noisy and seasonal data.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**25**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**15**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

1) **Problem**

A banking system wants to identify customers who are likely to default on their loans using financial history and transaction patterns.

Proposed Solution

K-Nearest Neighbour (KNN) can be used to classify customers as:

- Defaulter
- Non-defaulter

KNN is a distance-based algorithm. For a new customer, it finds the `K` most similar customers in the training dataset and assigns the class based on the majority of their classes.

Important Features

The following features can be selected:

- Income
- Credit Score
- Loan Amount
- Debt-to-Income Ratio
- Previous Loan Defaults
- Transaction Frequency
- Account Balance
- Employment Duration

Features should be scaled because KNN uses distance calculations.

Similarity Measurement

Euclidean distance can be used:

$$d(x,y)=\sqrt{\sum_{i=1}^{n}(x_i-y_i)^2}$$

For example, if a new customer has financial characteristics similar to five existing customers and four of them are non-defaulters, KNN predicts the new customer as a **non-defaulter**.

Working

1. Collect historical customer data.
2. Clean missing and incorrect values.
3. Select important financial features.
4. Normalize the features.
5. Select a suitable value of K.
6. Calculate distance between the new customer and existing customers.

7. Select the K nearest customers.
8. Use majority voting to predict the class.
9. Continuously update the dataset with new customer records.

Adapting to New Customers

One advantage of KNN is that it does not require complete retraining whenever new labeled customer data becomes available. New examples can be added to the database and used for future predictions.

For example:

```
New Customer
     ↓
Calculate similarity
     ↓
Find K nearest customers
     ↓
Majority voting
     ↓
Loan Default / No Default
```

Advantages

- Simple to implement.
- No complex training process.
- Can adapt to new examples.
- Works well when similar customers have similar outcomes.

Limitations

- Prediction can be slow for very large datasets.
- Sensitive to feature scaling.
- Choice of K affects performance.
- Irrelevant features can reduce accuracy.
- Outliers can affect the prediction.

Conclusion

KNN can provide a useful loan-default prediction system by identifying customers with similar financial behavior. Proper feature selection, normalization, and selection of K are important for accurate predictions. The system can continuously incorporate new customer data to improve future decisions.

2) Problem

A cybersecurity firm wants to detect abnormal network behavior and identify possible cyberattacks.

Proposed Solution

K-Nearest Neighbour (KNN) can classify network traffic as:

- Normal
- Intrusion

The system compares a new network connection with previously observed network patterns.

Network Features

Important features can include:

- Source IP
- Destination IP
- Protocol
- Number of packets
- Connection duration
- Bytes sent
- Bytes received
- Number of failed connections
- Port number
- Packet rate

Similarity Measurement

Euclidean distance can be used for numerical features:

$$
d(x,y)=\sqrt{\sum_{i=1}^{n}(x_i-y_i)^2}
$$

Before calculating distance, numerical features should be normalized so that features with large values do not dominate the distance.

Working

```
Network Traffic
      ↓
Feature Extraction
      ↓
Feature Scaling
      ↓
Calculate Similarity
      ↓
Find K Nearest Patterns
      ↓
Majority Voting
      ↓
Normal / Intrusion
```

For example, if a new network connection is similar to previously recorded attack patterns, the KNN classifier can identify it as an intrusion.

Anomaly Detection

If most nearby historical patterns belong to attack classes, the new connection is classified as suspicious.

For example:

Nearest Patterns	Classification
Attack	Attack
Attack	Attack
Attack	Attack
Normal	Normal
Attack	Attack

With `K = 5`, the majority is **Attack**, so the new traffic is classified as an intrusion.

Case-Based Learning

Case-Based Learning stores previous attack cases and uses them to solve new problems.

When a new attack pattern occurs:

1. Search for similar historical cases.
2. Compare their characteristics.
3. Identify the closest attack pattern.
4. Classify the new event.
5. Store the new confirmed attack as a future case.

Updating the Model

New attack patterns can be added to the case database.

For example:

```
Old Attack Cases
       +
New Confirmed Attack
       ↓
Updated Case Database
       ↓
Better Future Detection
```

This is useful because cyberattacks continuously evolve.

Advantages

- Detects patterns similar to known attacks.
- Easy to update with new cases.
- Does not require complex model training.

- Can classify different types of network behavior.

Limitations

- Large case databases can increase prediction time.
- Unknown attacks may be difficult to detect.
- Noise and irrelevant features can affect similarity.
- Choosing an appropriate K is important.

Conclusion

KNN and Case-Based Learning are useful for intrusion detection because they compare new network behavior with known patterns. Case-Based Learning is particularly useful in cybersecurity because newly confirmed attacks can be stored and reused to identify similar future attacks.

3) Problem

An energy management company wants to predict electricity consumption using factors such as time, weather, and appliance usage.

Proposed Solution

Locally Weighted Regression (LWR) can be used to predict electricity consumption. Unlike ordinary regression, LWR gives greater importance to data points that are close to the current input.

Input Features

The system can use:

- Hour of the day
- Day of the week
- Temperature
- Humidity
- Previous electricity consumption
- Number of appliances being used
- Season
- Holiday/weekend information

The output is:

Predicted electricity consumption

Local Learning

LWR calculates a weight for every training example based on its distance from the new input.

A commonly used weight function is:

$$
w_i=e^{-\frac{(x-x_i)^2}{2\tau^2}}
$$

where:

- (x) = current input
- (x_i) = training data point
- (τ) = bandwidth parameter

Nearby observations receive higher weights, while distant observations receive lower weights.

Working

```
Historical Energy Data
        ↓
Feature Extraction
        ↓
Time + Weather + Appliance Data
        ↓
Calculate Local Weights
        ↓
Fit Local Regression
        ↓
Predict Consumption
```

Example

Suppose electricity consumption is being predicted at **7 PM**.

Data points around 6 PM, 7 PM, and 8 PM will receive higher weights because they are close to the target time.

Data from very different periods will receive lower weights.

Therefore, the model can capture local variations in electricity usage.

Handling Seasonal Data

Electricity usage changes with seasons.

For example:

- Summer → Higher cooling usage
- Winter → Different heating requirements
- Weekends → Different usage patterns
- Working days → Different consumption patterns

Time and seasonal features can be included in the model to capture these variations.

Handling Noise

Real-world electricity data may contain abnormal readings caused by:

- Sensor errors
- Sudden power consumption
- Missing values
- Unexpected appliance usage

Data preprocessing, normalization and removal/imputation of abnormal values can reduce the effect of noise.

RBF Alternative

A **Radial Basis Function (RBF) network** can also model nonlinear electricity consumption.

The RBF function is commonly:

$$
\phi(x)=e^{-\frac{||x-c||^2}{2\sigma^2}}
$$

where:

- (x) = input
- (c) = center
- (σ) = spread

RBF networks are useful when electricity consumption has complex nonlinear relationships with weather and appliance usage.

Advantages

- Captures local consumption patterns.
- Can model nonlinear relationships.
- Adapts to changing patterns.
- Useful for time and weather-dependent prediction.

Limitations

- LWR can become computationally expensive for large datasets.
- Choice of bandwidth affects prediction.
- Noisy data can reduce accuracy.
- RBF networks require suitable selection of centers and spread.

Conclusion

Locally Weighted Regression is suitable for electricity consumption prediction because it gives more importance to nearby and relevant observations. It can capture local changes caused by time, weather and appliance usage. RBF networks provide another effective solution when the relationship between input variables and electricity consumption is highly nonlinear.